Run report: teske_variants_field_types, run 20261018T0630Z

Descriptive record of one run of `examples/teske_variants_field_types.rs` (protocol `research/teske_variants_field_types_20261018/PROTOCOL.md`, frozen at commit `62b4502c7`). It lists what ran, on what host, and the values the tool recorded, with their files. Interpretation is in `REPORT.md` of the study. Nothing here is a running time of an attack; the only timings are the tool's own wall-clock seconds per part, from `tool_log.txt`.

Command, host and isolation

```
run: 20261018T0630Z
protocol commit: 62b4502c7
working tree: 62b4502c7 plus the uncommitted tool (committed with the results)
rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v139 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.80GHz, 15 GB
command: rustc --edition 2021 -O examples/teske_variants_field_types.rs -o tvft && ./tvft research/
teske_variants_field_types_20261018/results/run-20261018T0630Z A B C
tool sha256: 437ebf9244ad1b8b42242531fea097588b4b310b326515c80110a35ae96c1435
threads: 4 (std::thread::scope)

note 1: the tool file was reformatted with rustfmt after this run started (whitespace only; sha256 of the formatted
file at that point 4dd68f202b057ffd...); Part A output of the formatted binary was compared with these files and is
identical. The committed file further replaces the census root search (see the next run directory), which does not
touch Parts A and B.
note 2: the session worker process was restarted while Part C was at q = 13 and this tool process was killed. Parts A
and B are complete here (29 part lines in tool_log.txt, every (q, n) of the protocol); the census files hold q = 3,
5, 7, 11 and are superseded by the next run directory, whose q <= 11 census files are byte-identical to these.
```

Isolation: none earned (a shared cloud container, 4 threads, no pinning); the tool is exact, so isolation affects only the wall-clock figures below.

Tool log (wall-clock per part)

```
part A done
elapsed 0.0s
part B q=3 n=2 L=6 types=5 enum=0.0s traces=0.0s all_j=true
part B q=5 n=2 L=6 types=5 enum=0.0s traces=0.0s all_j=true
part B q=7 n=2 L=6 types=5 enum=0.0s traces=0.0s all_j=true
part B q=11 n=2 L=6 types=5 enum=0.4s traces=0.0s all_j=true
part B q=13 n=2 L=6 types=5 enum=1.2s traces=0.0s all_j=true
part B q=3 n=3 L=6 types=7 enum=0.0s traces=0.0s all_j=true
part B q=5 n=3 L=6 types=7 enum=0.0s traces=0.0s all_j=true
part B q=7 n=3 L=6 types=7 enum=0.0s traces=0.0s all_j=true
part B q=11 n=3 L=6 types=7 enum=0.8s traces=0.0s all_j=true
part B q=13 n=3 L=6 types=7 enum=2.5s traces=0.1s all_j=true
part B q=3 n=4 L=4 types=3 enum=0.0s traces=0.0s all_j=true
part B q=5 n=4 L=4 types=3 enum=0.0s traces=0.0s all_j=true
part B q=7 n=4 L=4 types=3 enum=0.0s traces=0.0s all_j=true
part B q=11 n=4 L=4 types=3 enum=0.3s traces=1.4s all_j=true
part B q=13 n=4 L=4 types=3 enum=0.9s traces=3.3s all_j=true
part B q=3 n=5 L=5 types=1 enum=0.0s traces=0.0s all_j=true
part B q=5 n=5 L=5 types=1 enum=0.0s traces=0.0s all_j=true
part B q=7 n=5 L=5 types=1 enum=0.3s traces=1.1s all_j=true
part B q=11 n=5 L=5 types=1 enum=6.8s traces=111.8s all_j=true
part B q=13 n=5 L=5 types=1 enum=19.3s traces=645.0s all_j=true
part B q=3 n=6 L=6 types=2 enum=0.0s traces=0.0s all_j=true
part B q=5 n=6 L=6 types=2 enum=0.0s traces=1.0s all_j=true
part B q=7 n=6 L=6 types=2 enum=0.1s traces=58.8s all_j=true
part B q=11 n=6 L=6 types=2 enum=1.0s traces=26.8s all_j=false
part B q=13 n=6 L=6 types=2 enum=2.6s traces=182.6s all_j=false
part B q=3 n=7 L=7 types=1 enum=0.0s traces=0.0s all_j=true
part B q=5 n=7 L=7 types=1 enum=0.0s traces=25.6s all_j=true
part B q=7 n=7 L=7 types=1 enum=0.4s traces=19.8s all_j=false
elapsed 1114.5s
```

Part A: binary fields (`binary_types.csv`, `binary_gghs.csv`, `binary_summary.csv`, `binary_bits.csv`)

Rows in `binary_types.csv`: 766; in `binary_gghs.csv`: 541; in `binary_bits.csv`: 164. Every value is an exact count or a formula on the cyclotomic cosets of 2 modulo n; the per-class expectations are the GHS 2002 §5 formula. Selected rows of the summary (smallest genus per variant; δ = n − deg F; margin a = δ − n/2):

n	ord	cosets	type I (deg,g,δ)	a_I	type II (deg,g,δ)	a_II	gGHS (r,s,g,δ)	a_gGHS
3	2	2	(2, 3, 1)	−0.5	(3, 4, 0)	−1.5	(2, 2, 3, 0)	−1.5
5	4	2	(4, 15, 1)	−1.5	(5, 16, 0)	−2.5	(1, 4, 15, 1)	−1.5
7	3	3	(3, 7, 4)	0.5	(4, 8, 3)	−0.5	(3, 3, 7, 2)	−1.5
9	6	3	(6, 63, 3)	−1.5	(7, 64, 2)	−2.5	(1, 6, 63, 3)	−1.5
15	4	5	(4, 15, 11)	3.5	(5, 16, 10)	2.5	(4, 4, 15, 8)	0.5
17	8	3	(8, 255, 9)	0.5	(9, 256, 8)	−0.5	(8, 8, 255, 2)	−6.5
21	6	6	(5, 31, 16)	5.5	(6, 32, 15)	4.5	(2, 3, 21, 17)	6.5
23	11	3	(11, 2047, 12)	0.5	(12, 2048, 11)	−0.5	(11, 11, 2047, 2)	−9.5
31	5	7	(5, 31, 26)	10.5	(6, 32, 25)	9.5	(5, 5, 31, 22)	6.5
33	10	5	(10, 1023, 23)	6.5	(11, 1024, 22)	5.5	(10, 10, 1023, 14)	−2.5

35	12	6	(7, 127, 28)	10.5	(8, 128, 27)	9.5	(3, 4, 105, 29)	11.5
45	12	8	(10, 1023, 35)	12.5	(11, 1024, 34)	11.5	(4, 6, 945, 36)	13.5
63	6	13	(6, 63, 57)	25.5	(7, 64, 56)	24.5	(6, 6, 63, 52)	20.5
93	10	14	(7, 127, 86)	39.5	(8, 128, 85)	38.5	(2, 5, 93, 87)	40.5

Bit table rows for N = 155 and 161 (all variants except `generic`):

N	n	l	variant	g	δ	rho	owner	walk	margin	log2 #weak	per class
155	5	31	GHS-I	15	1	77.5	57.9	31.0	−46.5	125.0	1.000e0
155	5	31	GHS-II	16	0	77.5	58.1	0.0	−77.5	156.0	1.000e0
155	5	31	gGHS	15	1	77.5	57.9	31.0	−46.5	126.0	1.000e0
155	31	5	GHS-I	31	26	77.5	9.7	77.5	52.5	28.6	6.078e-18
155	31	5	GHS-II	32	25	77.5	9.7	77.5	47.5	33.6	1.945e-16
155	31	5	gGHS	31	22	77.5	9.7	77.5	32.5	48.6	6.373e-12
161	7	23	GHS-I	7	4	80.5	39.4	80.5	11.5	71.0	4.289e-6
161	7	23	GHS-II	8	3	80.5	40.2	69.0	−11.5	94.0	1.000e0
161	7	23	gGHS	7	2	80.5	39.4	46.0	−34.5	117.0	1.000e0
161	23	7	GHS-I	2047	12	80.5	14.0	80.5	3.5	79.0	1.098e-3
161	23	7	GHS-II	2048	11	80.5	14.0	77.0	−3.5	86.0	1.405e-1
161	23	7	gGHS	2047	2	80.5	14.0	14.0	−66.5	149.0	1.000e0

Part B: odd characteristic (`oddchar_types.csv`, `oddchar_classes.csv`, `oddchar_checks.txt`)

Rows (one per (q, n, branch type)): 93. instances_B is the number of weak branch sets (pairs (E, x)); weak_j the number of distinct j-invariants; ratio = weak_j / q^(r−3).

q	n	L	orbits	r	m	g	instances B	weak j	log_q	pred.	ratio	all-j
3	2	6	2+1+1+1	5	2	2	24	1	0.0000	2	0.1111	true
3	2	6	2+2+1	5	2	2	48	1	0.0000	2	0.1111	true
3	2	6	2+2+1+1	6	2	3	72	2	0.6309	3	0.0741	true
3	2	6	2+2+2	6	2	3	12	1	0.0000	3	0.0370	true
3	2	6	3+2	5	2	2	48	2	0.6309	2	0.2222	true
5	2	6	2+1+1+1	5	2	2	400	4	0.8614	2	0.1600	true
5	2	6	2+2+1	5	2	2	1080	4	0.8614	2	0.1600	true
5	2	6	2+2+1+1	6	2	3	2700	5	1.0000	3	0.0400	true
5	2	6	2+2+2	6	2	3	1440	5	1.0000	3	0.0400	true
5	2	6	3+2	5	2	2	800	7	1.2091	2	0.2800	true
7	2	6	2+1+1+1	5	2	2	2352	7	1.0000	2	0.1429	true
7	2	6	2+2+1	5	2	2	6720	9	1.1292	2	0.1837	true
7	2	6	2+2+1+1	6	2	3	23520	9	1.1292	3	0.0262	true
7	2	6	2+2+2	6	2	3	15960	9	1.1292	3	0.0262	true
7	2	6	3+2	5	2	2	4704	14	1.3562	2	0.2857	true
11	2	6	2+1+1+1	5	2	2	24200	19	1.2279	2	0.1570	true
11	2	6	2+2+1	5	2	2	71280	20	1.2493	2	0.1653	true
11	2	6	2+2+1+1	6	2	3	392040	21	1.2697	3	0.0158	true
11	2	6	2+2+2	6	2	3	314820	21	1.2697	3	0.0158	true
11	2	6	3+2	5	2	2	48400	37	1.5059	2	0.3058	true
13	2	6	2+1+1+1	5	2	2	56784	26	1.2702	2	0.1538	true
13	2	6	2+2+1	5	2	2	168168	29	1.3128	2	0.1716	true
13	2	6	2+2+1+1	6	2	3	1093092	29	1.3128	3	0.0132	true
13	2	6	2+2+2	6	2	3	912912	29	1.3128	3	0.0132	true
13	2	6	3+2	5	2	2	113568	52	1.5405	2	0.3077	true
3	3	6	3+1+1	5	3	3	144	3	1.0000	2	0.3333	true
3	3	6	3+1+1+1	6	3	5	96	4	1.2619	3	0.1481	true
3	3	6	3+2	5	3	3	72	3	1.0000	2	0.3333	true
3	3	6	3+2+1	6	3	5	288	12	2.2619	3	0.4444	true

3	3	6	3+3	6	2	3	252	4	1.2619	3	0.1481	true
3	3	6	3+3	6	3	5	168	4	1.2619	3	0.1481	true
3	3	6	6	6	2	3	348	5	1.4650	3	0.1852	true
5	3	6	3+1+1	5	3	3	1800	12	1.5440	2	0.4800	true
5	3	6	3+1+1+1	6	3	5	2400	20	1.8614	3	0.1600	true
5	3	6	3+2	5	3	3	1200	10	1.4307	2	0.4000	true
5	3	6	3+2+1	6	3	5	7200	60	2.5440	3	0.4800	true
5	3	6	3+3	6	2	3	7020	21	1.8917	3	0.1680	true
5	3	6	3+3	6	3	5	4680	21	1.8917	3	0.1680	true
5	3	6	6	6	2	3	7740	21	1.8917	3	0.1680	true
7	3	6	3+1+1	5	3	3	9408	25	1.6542	2	0.5102	true
7	3	6	3+1+1+1	6	3	5	18816	56	2.0686	3	0.1633	true
7	3	6	3+2	5	3	3	7056	21	1.5646	2	0.4286	true
7	3	6	3+2+1	6	3	5	56448	168	2.6332	3	0.4898	true
7	3	6	3+3	6	2	3	55944	58	2.0867	3	0.1691	true
7	3	6	3+3	6	3	5	37296	57	2.0777	3	0.1662	true
7	3	6	6	6	2	3	58632	58	2.0867	3	0.1691	true
11	3	6	3+1+1	5	3	3	87120	60	1.7075	2	0.4959	true
11	3	6	3+1+1+1	6	3	5	290400	220	2.2493	3	0.1653	true
11	3	6	3+2	5	3	3	72600	55	1.6712	2	0.4545	true
11	3	6	3+2+1	6	3	5	871200	660	2.7075	3	0.4959	true
11	3	6	3+3	6	2	3	869220	221	2.2512	3	0.1660	true
11	3	6	3+3	6	3	5	579480	222	2.2531	3	0.1668	true
11	3	6	6	6	2	3	885060	221	2.2512	3	0.1660	true
13	3	6	3+1+1	5	3	3	198744	82	1.7181	2	0.4852	true
13	3	6	3+1+1+1	6	3	5	794976	364	2.2991	3	0.1657	true
13	3	6	3+2	5	3	3	170352	78	1.6986	2	0.4615	true
13	3	6	3+2+1	6	3	5	2384928	1092	2.7274	3	0.4970	true
13	3	6	3+3	6	2	3	2381652	363	2.2981	3	0.1652	true
13	3	6	3+3	6	3	5	1587768	366	2.3013	3	0.1666	true
13	3	6	6	6	2	3	2412228	366	2.3013	3	0.1666	true
3	4	4	4+1	5	4	5	288	7	1.7712	2	0.7778	true
3	4	4	4+1+1	6	3	5	432	13	2.3347	3	0.4815	true
3	4	4	4+2	6	3	5	216	9	2.0000	3	0.3333	true
5	4	4	4+1	5	4	5	3600	27	2.0478	2	1.0800	true
5	4	4	4+1+1	6	3	5	9000	63	2.5743	3	0.5040	true
5	4	4	4+2	6	3	5	6000	50	2.4307	3	0.4000	true
7	4	4	4+1	5	4	5	18816	49	2.0000	2	1.0000	true
7	4	4	4+1+1	6	3	5	65856	171	2.6423	3	0.4985	true
7	4	4	4+2	6	3	5	49392	147	2.5646	3	0.4286	true
11	4	4	4+1	5	4	5	174240	126	2.0169	2	1.0413	true
11	4	4	4+1+1	6	3	5	958320	686	2.7236	3	0.5154	true
11	4	4	4+2	6	3	5	798600	605	2.6712	3	0.4545	true
13	4	4	4+1	5	4	5	397488	172	2.0069	2	1.0178	true
13	4	4	4+1+1	6	3	5	2583672	1110	2.7338	3	0.5052	true
13	4	4	4+2	6	3	5	2214576	1014	2.6986	3	0.4615	true
3	5	5	5	5	4	5	240	10	2.0959	2	1.1111	true
5	5	5	5	5	4	5	3120	26	2.0244	2	1.0400	true
7	5	5	5	5	4	5	16800	50	2.0104	2	1.0204	true
11	5	5	5	5	4	5	161040	121	2.0000	2	1.0000	true
13	5	5	5	5	4	5	371280	170	2.0023	2	1.0059	true
3	6	6	3+2+1	6	4	9	576	21	2.7712	3	0.7778	true

3	6	6	6	6	4	9	696	23	2.8540	3	0.8519	true
5	6	6	3+2+1	6	4	9	14400	117	2.9589	3	0.9360	true
5	6	6	6	6	4	9	15480	121	2.9798	3	0.9680	true
7	6	6	3+2+1	6	4	9	112896	336	2.9894	3	0.9796	true
7	6	6	6	6	4	9	117264	334	2.9863	3	0.9738	true
11	6	6	3+2+1	6	4	9	1742400	1320	2.9965	3	0.9917	false
11	6	6	6	6	4	9	1770120	1323	2.9975	3	0.9940	false
13	6	6	3+2+1	6	4	9	4769856	2184	2.9977	3	0.9941	false
13	6	6	6	6	4	9	4824456	2181	2.9972	3	0.9927	false
3	7	7	7	7	3	7	4368	147	4.5425	4	1.8148	true
5	7	7	7	7	3	7	156240	1253	4.4322	4	2.0048	true
7	7	7	7	7	3	7	1647072	4782	4.3541	4	1.9917	false

Class statistics (rows: 88; only where the all-j trace table was computed): classes is the number of traces realised over F_(q^n); covered are those with a weak member of the type; share is the fraction of all curves (both twists of every j) in covered classes; walk = curves in covered classes / weak curves in them.

q	n	orbits	r	m	g	classes	covered	share	mean weak	max weak	walk
3	2	2+1+1+1	5	2	2	10	2	0.3333	1.000	1	3.00
3	2	2+2+1	5	2	2	10	2	0.3333	1.000	1	3.00
3	2	2+2+1+1	6	2	3	10	4	0.4444	1.000	1	2.00
3	2	2+2+2	6	2	3	10	2	0.3333	1.000	1	3.00
3	2	3+2	5	2	2	10	2	0.2222	2.000	2	1.00
5	2	2+1+1+1	5	2	2	18	6	0.4400	1.333	2	2.75
5	2	2+2+1	5	2	2	18	4	0.4000	2.000	2	2.50
5	2	2+2+1+1	6	2	3	18	6	0.4400	1.667	2	2.20
5	2	2+2+2	6	2	3	18	6	0.4400	1.667	2	2.20
5	2	3+2	5	2	2	18	8	0.3200	1.750	2	1.14
7	2	2+1+1+1	5	2	2	26	6	0.4082	2.333	3	2.86
7	2	2+2+1	5	2	2	26	8	0.4286	2.250	4	2.33
7	2	2+2+1+1	6	2	3	26	8	0.4286	2.250	4	2.33
7	2	2+2+2	6	2	3	26	8	0.4286	2.250	4	2.33
7	2	3+2	5	2	2	26	10	0.3061	2.800	4	1.07
11	2	2+1+1+1	5	2	2	42	12	0.4215	3.167	6	2.68
11	2	2+2+1	5	2	2	42	12	0.4215	3.333	6	2.55
11	2	2+2+1+1	6	2	3	42	12	0.4215	3.500	6	2.43
11	2	2+2+2	6	2	3	42	12	0.4215	3.500	6	2.43
11	2	3+2	5	2	2	42	20	0.3388	3.700	8	1.11
13	2	2+1+1+1	5	2	2	50	14	0.4201	3.714	7	2.73
13	2	2+2+1	5	2	2	50	14	0.4201	4.143	8	2.45
13	2	2+2+1+1	6	2	3	50	14	0.4201	4.143	8	2.45
13	2	2+2+2	6	2	3	50	14	0.4201	4.143	8	2.45
13	2	3+2	5	2	2	50	22	0.3195	4.727	8	1.04
3	3	3+1+1	5	3	3	15	2	0.2222	3.000	3	2.00
3	3	3+1+1+1	6	3	5	15	4	0.3704	2.000	3	2.50
3	3	3+2	5	3	3	15	2	0.2222	3.000	3	2.00
3	3	3+2+1	6	3	5	15	6	0.5926	4.000	6	1.33
3	3	3+3	6	2	3	15	4	0.3704	2.000	3	2.50
3	3	3+3	6	3	5	15	4	0.3704	2.000	3	2.50
3	3	6	6	2	3	15	5	0.4074	2.000	3	2.20
5	3	3+1+1	5	3	3	37	6	0.2560	4.000	6	2.67
5	3	3+1+1+1	6	3	5	37	8	0.4000	5.000	6	2.50
5	3	3+2	5	3	3	37	8	0.3360	2.500	3	4.20
5	3	3+2+1	6	3	5	37	16	0.6400	7.500	12	1.33

5	3	3+3	6	2	3	37	10	0.4160	4.200	6	2.48
5	3	3+3	6	3	5	37	10	0.4160	4.200	6	2.48
5	3	6	6	2	3	37	10	0.4160	4.200	6	2.48
7	3	3+1+1	5	3	3	63	9	0.2478	5.556	12	3.40
7	3	3+1+1+1	6	3	5	63	16	0.4140	7.000	15	2.54
7	3	3+2	5	3	3	63	4	0.1224	10.500	12	2.00
7	3	3+2+1	6	3	5	63	32	0.6647	10.500	18	1.36
7	3	3+3	6	2	3	63	17	0.4169	6.824	15	2.47
7	3	3+3	6	3	5	63	17	0.4169	6.706	15	2.51
7	3	6	6	2	3	63	17	0.4169	6.824	15	2.47
11	3	3+1+1	5	3	3	133	16	0.2494	7.500	12	5.53
11	3	3+1+1+1	6	3	5	133	34	0.4147	12.941	36	2.51
11	3	3+2	5	3	3	133	16	0.2494	6.875	12	6.04
11	3	3+2+1	6	3	5	133	66	0.6649	20.000	48	1.34
11	3	3+3	6	2	3	133	35	0.4162	12.629	36	2.51
11	3	3+3	6	3	5	133	35	0.4162	12.686	36	2.50
11	3	6	6	2	3	133	35	0.4162	12.629	36	2.51
13	3	3+1+1	5	3	3	173	20	0.2230	8.200	18	5.98
13	3	3+1+1+1	6	3	5	173	42	0.4169	17.333	36	2.52
13	3	3+2	5	3	3	173	32	0.3414	4.875	6	9.62
13	3	3+2+1	6	3	5	173	86	0.6664	25.395	60	1.34
13	3	3+3	6	2	3	173	42	0.4169	17.286	33	2.52
13	3	3+3	6	3	5	173	42	0.4169	17.429	36	2.50
13	3	6	6	2	3	173	42	0.4169	17.429	36	2.50
3	4	4+1	5	4	5	26	4	0.2593	3.500	6	3.00
3	4	4+1+1	6	3	5	26	6	0.4074	4.333	6	2.54
3	4	4+2	6	3	5	26	4	0.2593	4.500	6	2.33
5	4	4+1	5	4	5	82	8	0.1792	6.750	9	4.15
5	4	4+1+1	6	3	5	82	10	0.2240	12.600	17	2.22
5	4	4+2	6	3	5	82	16	0.3328	6.250	12	4.16
7	4	4+1	5	4	5	170	14	0.1533	7.000	12	7.51
7	4	4+1+1	6	3	5	170	36	0.3865	9.500	17	5.43
7	4	4+2	6	3	5	170	22	0.2366	13.364	28	3.86
11	4	4+1	5	4	5	442	40	0.1790	6.300	14	20.80
11	4	4+1+1	6	3	5	442	94	0.3805	14.596	44	8.12
11	4	4+2	6	3	5	442	56	0.2322	21.607	48	5.62
13	4	4+1	5	4	5	626	40	0.1347	8.600	16	22.36
13	4	4+1+1	6	3	5	626	78	0.2276	28.462	60	5.86
13	4	4+2	6	3	5	626	144	0.4003	14.083	60	11.27
3	5	5	5	4	5	43	4	0.1646	5.000	5	4.00
5	5	5	5	4	5	179	10	0.1254	5.200	10	15.08
7	5	5	5	4	5	445	14	0.0768	7.143	15	25.80
11	5	5	5	4	5	1461	36	0.0608	6.722	20	80.93
13	5	5	5	4	5	2251	56	0.0539	6.071	10	117.76
3	6	3+2+1	6	4	9	74	8	0.2140	5.250	6	7.43
3	6	6	6	4	9	74	8	0.1742	5.750	7	5.52
5	6	3+2+1	6	4	9	402	24	0.1334	9.750	18	17.81
5	6	6	6	4	9	402	38	0.1841	6.368	12	23.77
7	6	3+2+1	6	4	9	1178	64	0.1195	10.500	30	41.83
7	6	6	6	4	9	1178	86	0.1414	7.767	30	49.82
3	7	7	7	3	7	125	24	0.3393	12.250	28	5.05
5	7	7	7	3	7	897	160	0.3362	15.662	42	20.96

Checks (`oddchar_checks.txt`):

```
q=3 n=2 weak_j_outside_K=0
q=3 n=2 all_j_table=true j_listed=9 hasse_violations=0
q=3 n=2 check6_num_j=9 expected=9
q=5 n=2 weak_j_outside_K=0
q=5 n=2 all_j_table=true j_listed=25 hasse_violations=0
q=5 n=2 check6_num_j=25 expected=25
q=7 n=2 weak_j_outside_K=0
q=7 n=2 all_j_table=true j_listed=49 hasse_violations=0
q=7 n=2 check6_num_j=49 expected=49
q=11 n=2 weak_j_outside_K=0
q=11 n=2 all_j_table=true j_listed=121 hasse_violations=0
q=11 n=2 check6_num_j=121 expected=121
q=13 n=2 weak_j_outside_K=0
q=13 n=2 all_j_table=true j_listed=169 hasse_violations=0
q=13 n=2 check6_num_j=169 expected=169
q=3 n=3 weak_j_outside_K=0
q=3 n=3 all_j_table=true j_listed=27 hasse_violations=0
q=3 n=3 check6_num_j=27 expected=27
q=3 n=3 type=3+1+1 m=3 r=5 check5_curves=3 check5_violations=0
q=3 n=3 type=3+2 m=3 r=5 check5_curves=3 check5_violations=0
q=5 n=3 weak_j_outside_K=0
q=5 n=3 all_j_table=true j_listed=125 hasse_violations=0
q=5 n=3 check6_num_j=125 expected=125
q=5 n=3 type=3+1+1 m=3 r=5 check5_curves=12 check5_violations=0
q=5 n=3 type=3+2 m=3 r=5 check5_curves=10 check5_violations=0
q=7 n=3 weak_j_outside_K=0
q=7 n=3 all_j_table=true j_listed=343 hasse_violations=0
q=7 n=3 check6_num_j=343 expected=343
q=7 n=3 type=3+1+1 m=3 r=5 check5_curves=24 check5_violations=0
q=7 n=3 type=3+2 m=3 r=5 check5_curves=21 check5_violations=0
q=11 n=3 weak_j_outside_K=0
q=11 n=3 all_j_table=true j_listed=1331 hasse_violations=0
q=11 n=3 check6_num_j=1331 expected=1331
q=11 n=3 type=3+1+1 m=3 r=5 check5_curves=60 check5_violations=0
q=11 n=3 type=3+2 m=3 r=5 check5_curves=55 check5_violations=0
q=13 n=3 weak_j_outside_K=0
q=13 n=3 all_j_table=true j_listed=2197 hasse_violations=0
q=13 n=3 check6_num_j=2197 expected=2197
q=13 n=3 type=3+1+1 m=3 r=5 check5_curves=82 check5_violations=0
q=13 n=3 type=3+2 m=3 r=5 check5_curves=78 check5_violations=0
q=3 n=4 weak_j_outside_K=0
q=3 n=4 all_j_table=true j_listed=81 hasse_violations=0
q=3 n=4 check6_num_j=81 expected=81
q=5 n=4 weak_j_outside_K=0
q=5 n=4 all_j_table=true j_listed=625 hasse_violations=0
q=5 n=4 check6_num_j=625 expected=625
q=7 n=4 weak_j_outside_K=0
q=7 n=4 all_j_table=true j_listed=2401 hasse_violations=0
q=7 n=4 check6_num_j=2401 expected=2401
q=11 n=4 weak_j_outside_K=0
q=11 n=4 all_j_table=true j_listed=14641 hasse_violations=0
q=11 n=4 check6_num_j=14641 expected=14641
q=13 n=4 weak_j_outside_K=0
q=13 n=4 all_j_table=true j_listed=28561 hasse_violations=0
q=13 n=4 check6_num_j=28561 expected=28561
q=3 n=5 weak_j_outside_K=0
q=3 n=5 all_j_table=true j_listed=243 hasse_violations=0
q=3 n=5 check6_num_j=243 expected=243
q=5 n=5 weak_j_outside_K=0
q=5 n=5 all_j_table=true j_listed=3125 hasse_violations=0
q=5 n=5 check6_num_j=3125 expected=3125
q=7 n=5 weak_j_outside_K=0
q=7 n=5 all_j_table=true j_listed=16807 hasse_violations=0
q=7 n=5 check6_num_j=16807 expected=16807
q=11 n=5 weak_j_outside_K=0
q=11 n=5 all_j_table=true j_listed=161051 hasse_violations=0
q=11 n=5 check6_num_j=161051 expected=161051
q=13 n=5 weak_j_outside_K=0
q=13 n=5 all_j_table=true j_listed=371293 hasse_violations=0
q=13 n=5 check6_num_j=371293 expected=371293
q=3 n=6 weak_j_outside_K=0
q=3 n=6 all_j_table=true j_listed=729 hasse_violations=0
q=3 n=6 check6_num_j=729 expected=729
q=5 n=6 weak_j_outside_K=0
q=5 n=6 all_j_table=true j_listed=15625 hasse_violations=0
q=5 n=6 check6_num_j=15625 expected=15625
q=7 n=6 weak_j_outside_K=0
q=7 n=6 all_j_table=true j_listed=117649 hasse_violations=0
q=7 n=6 check6_num_j=117649 expected=117649
q=11 n=6 weak_j_outside_K=0
q=11 n=6 all_j_table=false j_listed=2637 hasse_violations=0
q=13 n=6 weak_j_outside_K=0
```

```
q=13 n=6 all_j_table=false j_listed=4365 hasse_violations=0
q=3 n=7 weak_j_outside_K=0
q=3 n=7 all_j_table=true j_listed=2187 hasse_violations=0
q=3 n=7 check6_num_j=2187 expected=2187
q=5 n=7 weak_j_outside_K=0
q=5 n=7 all_j_table=true j_listed=78125 hasse_violations=0
q=5 n=7 check6_num_j=78125 expected=78125
q=7 n=7 weak_j_outside_K=0
q=7 n=7 all_j_table=false j_listed=4782 hasse_violations=0
```

Part C: census (`census.csv`, `census_summary.csv`, `census_patterns.csv`, `census_checks.txt`)

q	models	square-free	Res-shaped models	classes	signa-ture	no sig-nature	roots outside F_q^6	#E prime	prime, no sig.	t sub-field-shaped	of which sig.	traces	traces with sig.	s
3	5832	3888	152	21	6	15	0	0	0	7	0	9	2	0.0
5	187500	150000	3088	158	20	138	0	0	0	70	8	21	10	0.2
7	3294172	2823576	48352	538	39	499	0	0	0	92	1	37	9	5.5
11	42517464	38652240	293440	3173	106	3067	0	0	0	547	6	73	16	717.2

Classes by Weierstrass cycle type and signature:

q	cycle type	signature	classes
3	1+1+3+3	no	2
3	1+1+3+3	yes	4
3	1+1+6	no	11
3	2+6	no	2
3	2+6	yes	2
5	1+1+3+3	no	30
5	1+1+3+3	yes	12
5	1+1+6	no	76
5	2+3+3	no	12
5	2+6	no	20
5	2+6	yes	8
7	1+1+3+3	no	116
7	1+1+3+3	yes	19
7	1+1+6	no	211
7	2+3+3	no	38
7	2+6	no	134
7	2+6	yes	20
11	1+1+3+3	no	832
11	1+1+3+3	yes	54
11	1+1+6	no	1168
11	2+3+3	no	264
11	2+6	no	803
11	2+6	yes	52

Checks (`census_checks.txt`):

```
q=3 models=5832 squarefree=3888 res_shaped_models=152 classes=21 neighbours_missing=0
q=3 check8_hasse_violations=0 t_inconsistent_in_class=0 check9_signature_numE_mod4_violations=0
check9_signature_cycle_violations=0
q=5 models=187500 squarefree=150000 res_shaped_models=3088 classes=158 neighbours_missing=0
q=5 check8_hasse_violations=0 t_inconsistent_in_class=0 check9_signature_numE_mod4_violations=0
check9_signature_cycle_violations=0
q=7 models=3294172 squarefree=2823576 res_shaped_models=48352 classes=538 neighbours_missing=0
q=7 check8_hasse_violations=0 t_inconsistent_in_class=0 check9_signature_numE_mod4_violations=0
check9_signature_cycle_violations=0
q=11 models=42517464 squarefree=38652240 res_shaped_models=293440 classes=3173 neighbours_missing=0
q=11 check8_hasse_violations=0 t_inconsistent_in_class=0 check9_signature_numE_mod4_violations=0
check9_signature_cycle_violations=0
```

Graphs

Recorded values only, drawn by `figures/plot.py` from the CSVs of the two recorded runs (Parts A and B from run 20261018T0630Z, Part C from run 20261018T0845Z).

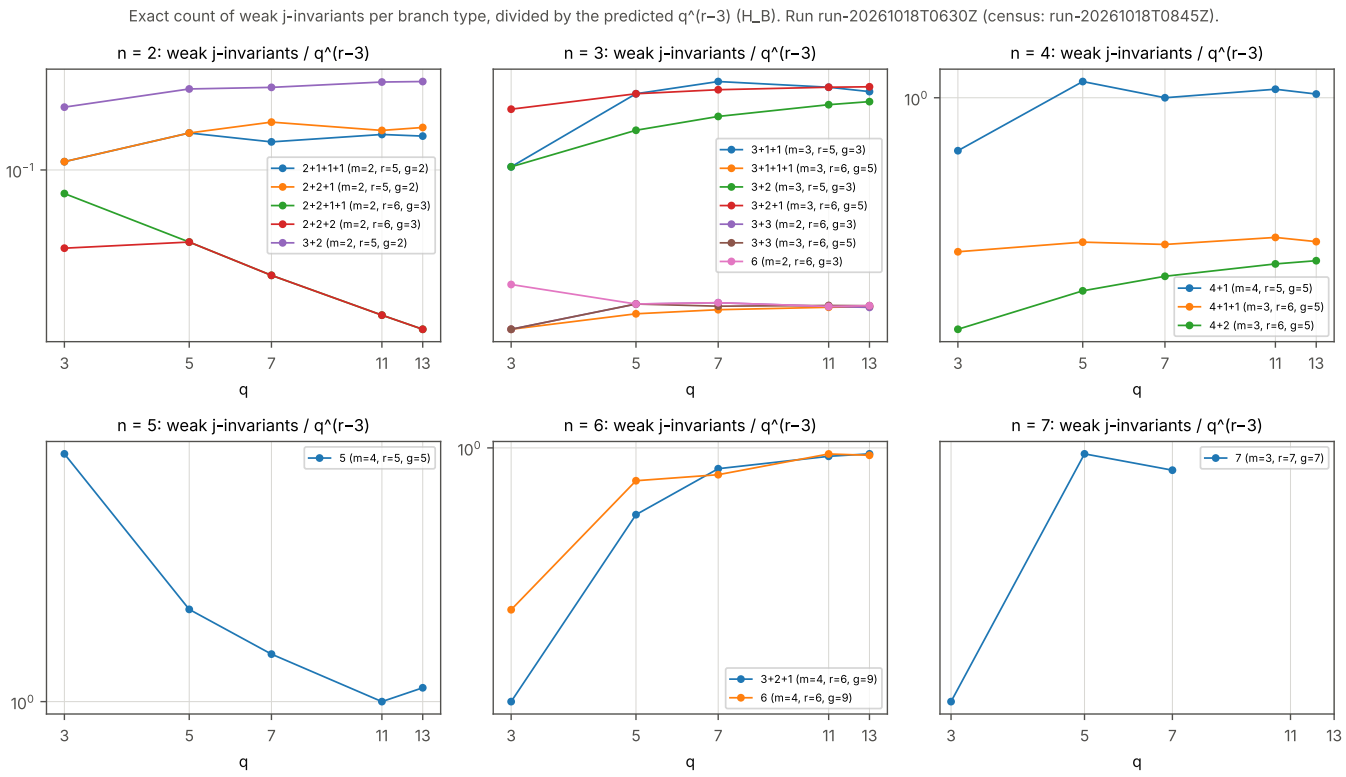


Figure 1: Weak j-invariants per branch type divided by q^(r−3); exact counts, no uncertainty.

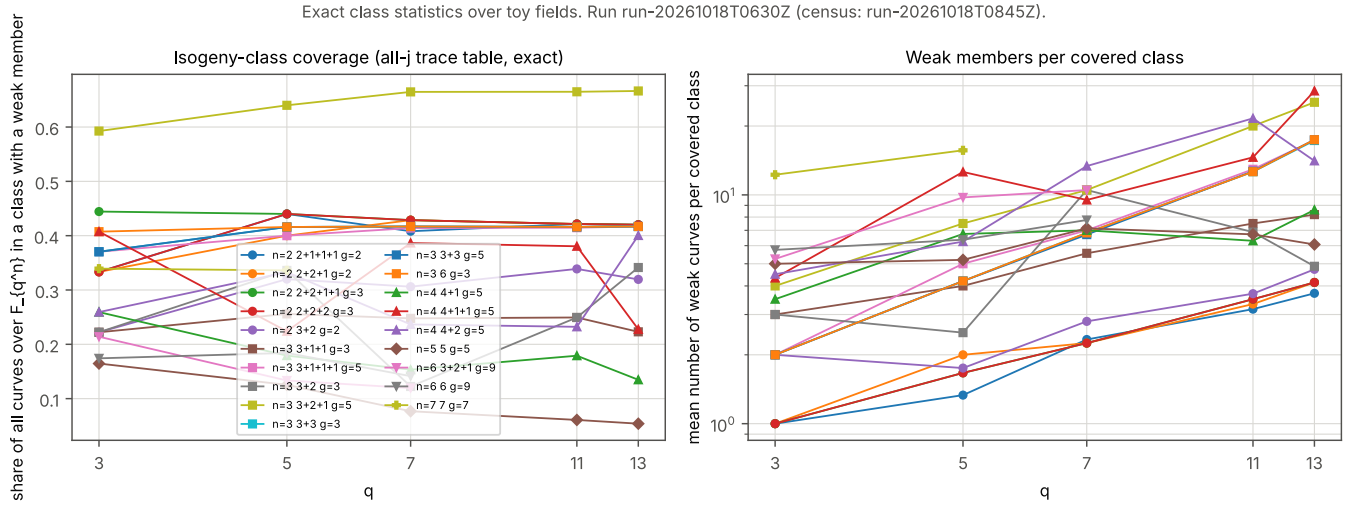


Figure 2: Share of curves in isogeny classes with a weak member, and weak members per covered class; exact, from the all-j trace tables.

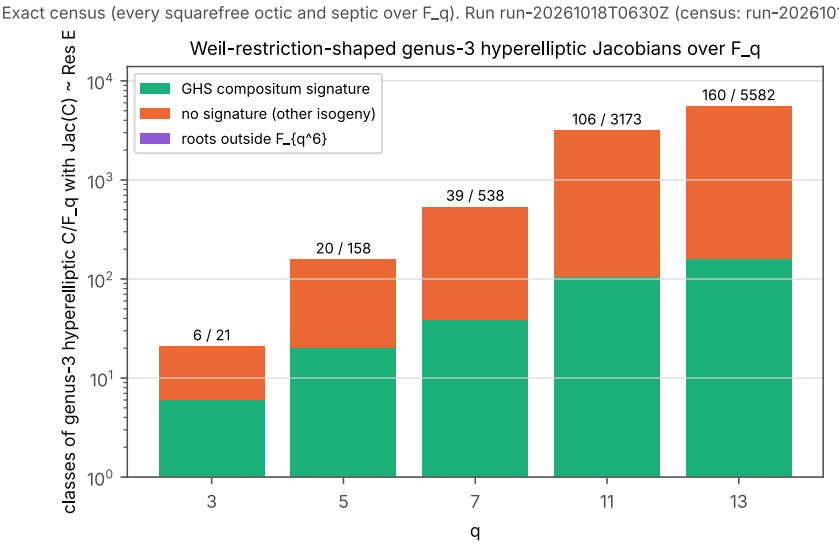


Figure 3: Res-shaped genus-3 hyperelliptic classes over F_q by GHS signature; exact census.

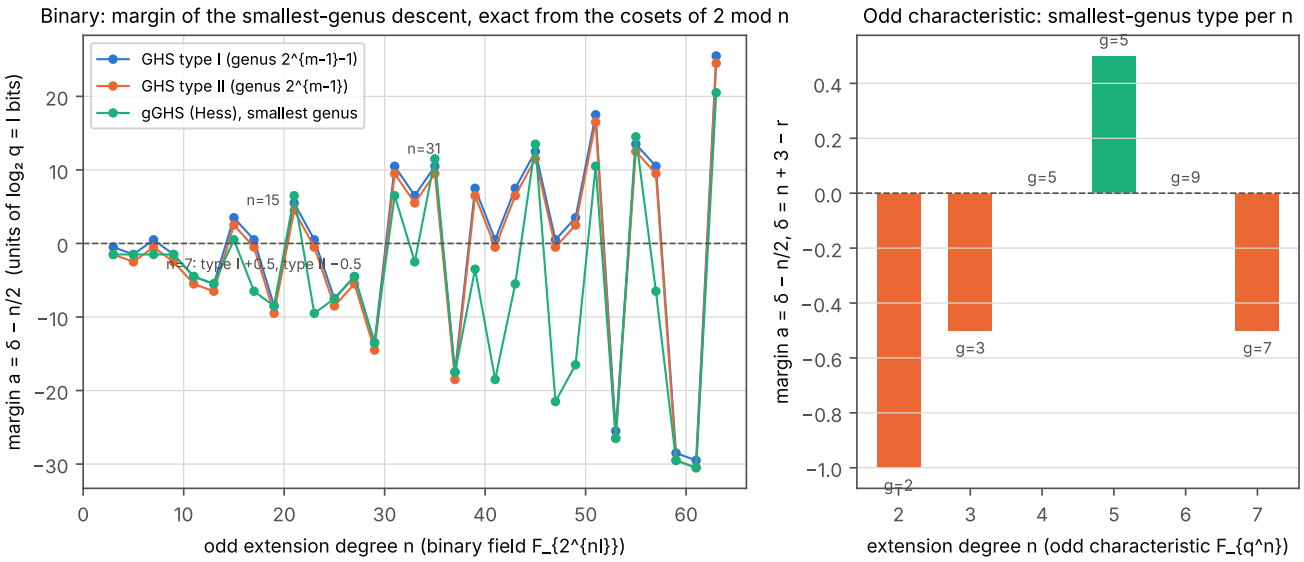


Figure 4: Margins a = δ − n/2 by n: binary (exact cosets) and odd characteristic (smallest-genus type).